

Statistical Analysis Report – Use Case Diagram Compilation

Hypotheses

This study investigates whether the domain-specific specialization of a language model affects its ability to generate syntactically valid use case diagrams.

H_0 (Null Hypothesis): The proportion of compilable diagrams is the same for both the specialized and non-specialized models ($p_{Fin} = p_{Qwen}$).

H_1 (Alternative Hypothesis): There is a statistically significant difference in the proportion of compilable diagrams between the two models ($p_{Fin} \neq p_{Qwen}$).

Descriptive Statistics

Total number of diagrams analyzed: 2000

Compiled successfully: 1519 diagrams

Failed to compile: 481 diagrams

These values are essential to understand the baseline distribution before statistical comparison.

Model	Compiled	Total	Proportion
Fin-R1-c1-latest	680	1000	0.6800
Qwen-Coder-c1-latest	839	1000	0.8390

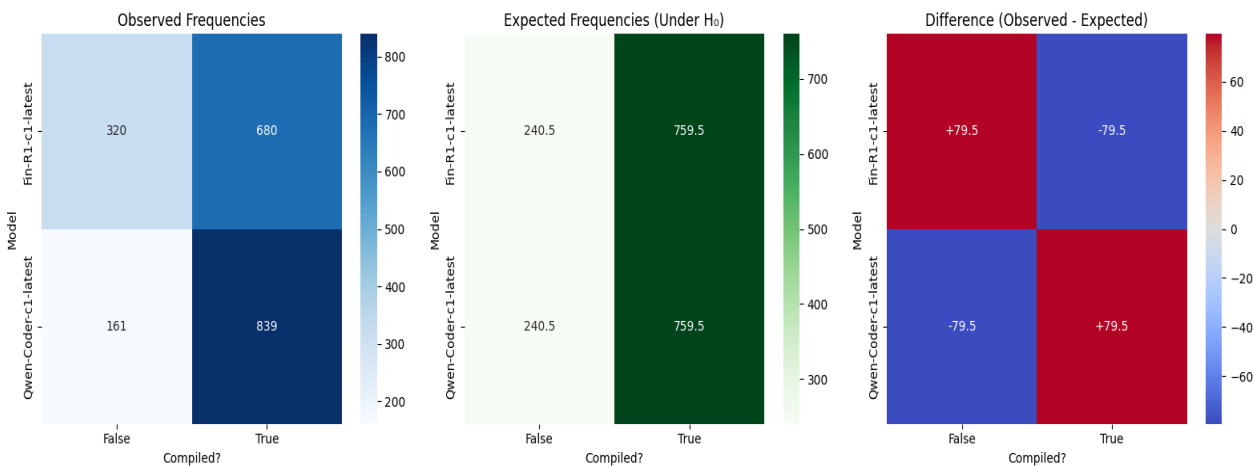
Z-Test Result

Z statistic: -8.3188

p-value: 8.8858e-17

The result is statistically significant ($p < 0.05$), indicating that the specialization impacts the model's performance in a measurable way.

Visual Analysis of Frequencies



Interpretation of Visual Frequencies

Model	Observation	Expected	Difference
Fin-R1-c1-latest - False	320	240.5	+79.5
Fin-R1-c1-latest - True	680	759.5	-79.5
Qwen-Coder-c1-latest - False	161	240.5	-79.5
Qwen-Coder-c1-latest - True	839	759.5	+79.5

Key observations reveal that the specialized model underperformed relative to expectations, while the non-specialized model exceeded expectations, suggesting a negative effect of specialization on syntactic validity.

Chi-Square Test Result

Chi² statistic: 68.3347

p-value: 1.3797e-16

This test confirms a statistically significant association between the type of model and the likelihood of generating compilable diagrams. It reinforces the Z-Test result by highlighting the dependency between model type and syntax success.

Integrated Conclusion

The non-specialized model (Qwen-Coder-c1-latest) compiled more diagrams correctly. This contradicts the expected advantage of specialization, suggesting that the added complexity may have introduced noise or rigid patterns.